PROOF OF CONCEPT · PROPOSAL FOR COMPANY A

Reach them before they decide to leave

A model-targeted handset-upgrade programme to reduce customer churn

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EXECUTIVE SUMMARY

Churn clusters at one predictable moment, which gives us a window to act

65%

THE MOMENT

The observed churn-label rate is 65% at month 11, compared with 29% at month 9. Contract dates are unavailable, so the reason for this increase remains uncertain.

83%

THE TRIGGER

83% of the highest-risk 10% have handsets older than 10 months. This association suggests an opportunity to test a targeted upgrade offer..

0.699

THE MODEL

ROC-AUC of LightGBM on 30,000 unseen customers. Its riskiest 10% churn at 80% in the sample.

+$0.20 M

THE PLAN

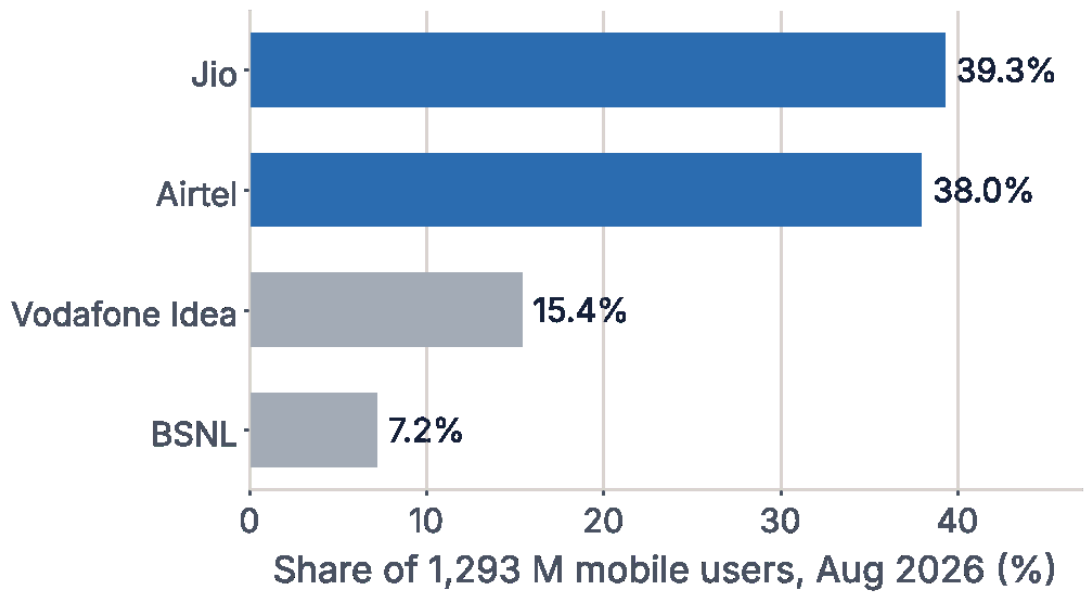
base case, 12 monthly campaigns per 1 M subscribers, for a targeted upgrade offer. To be confirmed by a pilot.

Company A data (n = 100,000). Impact depends on assumptions: −$0.28M to +$0.57M (slide 13)

REFERENCE MARKET · INDIA

India's Telecom Market Is Competitive, and Smartphone Financing Is Widespread

Two operators hold 77% of 1.29 billion mobile users [1]



Monthly revenue per user ranges from Rs 177 to Rs 264 [3], postpaid is growing fast [4], and analysts expect another 10–20% tariff rise [7].

14.5 M

Mobile-number portability requests submitted in India in May 2026. Requests do not necessarily represent completed switches. [2]

42%

Smartphone financing is forecast to account for 42% of smartphone sales in India in 2026. [5]

10 months

average EMI tenure for a smartphone (Q2 2026) [6]

5–25×

HBR's cited range for how much more it costs to win a customer than keep one [8]

India = reference context only. [1][2] TRAI · [3][4] Q1 FY27 results · [5][6] Counterpoint · [7][8]

COMPANY A'S DATA

A clean starting point, with two things to handle carefully

100,000

customers, 100 columns

Profile and usage/billing tables joined one-to-one on customer ID. No duplicates, no unmatched IDs.

49.6%

churned in this file

Far above any real operator; the sampling method is not documented. The business case assumes ~1.1% monthly churn instead [2].

76%

churn when usage history is missing

Churn-label rate among customers with missing usage-change values (n = 891). This is an observed association in the dataset, not evidence that missing data causes churn..

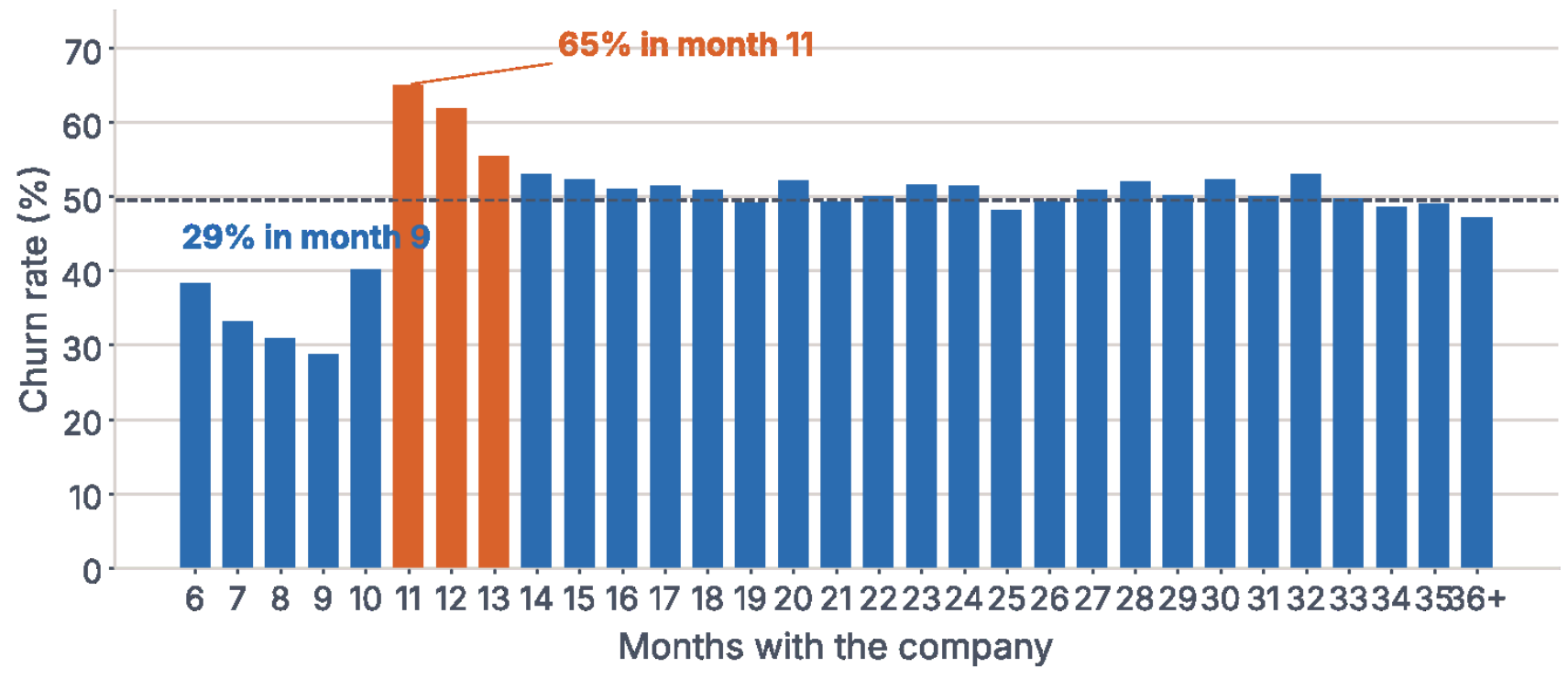
Also cleaned: 133 handset ages of −1 to −5 days set to 0 · 3 billing-adjusted copies removed (correlation ≥ 0.998) · demographic fields are 25–49% empty, so the proposal does not rely on them.

Source: Company A data (Client.csv, Record.csv); [2] TRAI, May 2026

FINDING 1 · TIMING

Churn Peaks Around Month 11 of Customer Tenure

Churn-label rate by months with Company A. The dashed line shows the overall churn-label prevalence in the sample (49.6%). The clearest pattern is a sharp rise in the churn-label rate around month 11. The rate reaches approximately 65%, compared with 29% at month 9. However, the dataset does not include actual contract dates, so we cannot confirm that contract expiry explains this increase. Months 910 are therefore a possible window for early retention activity, which should be tested in the pilot.



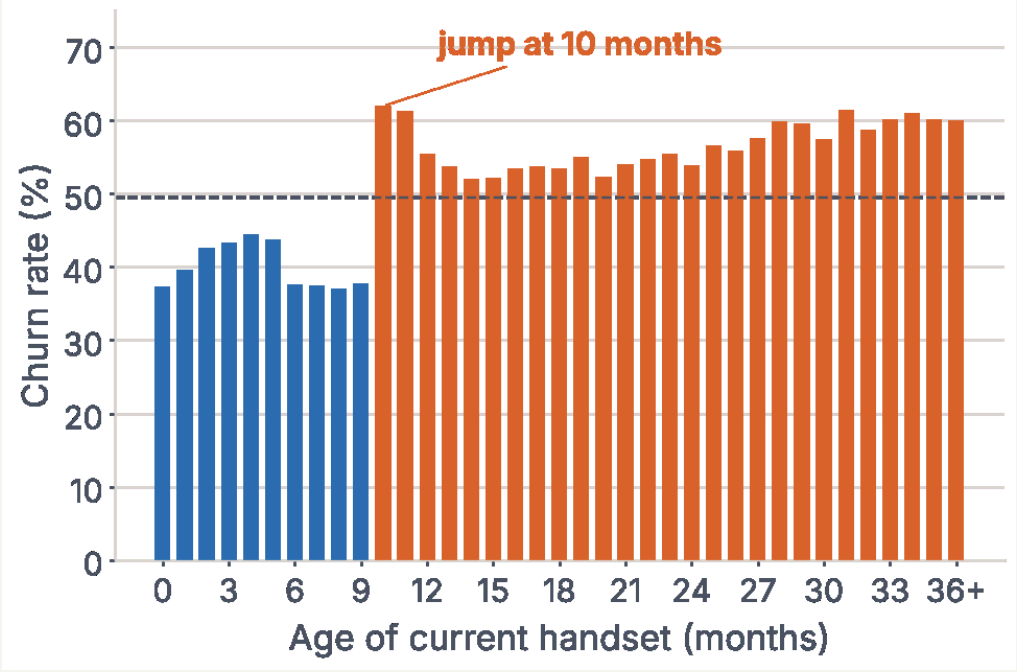
Source: Company A data, n = 100,000. Churn = left 31–60 days after the observation date.

FINDING 2 · THE HANDSET

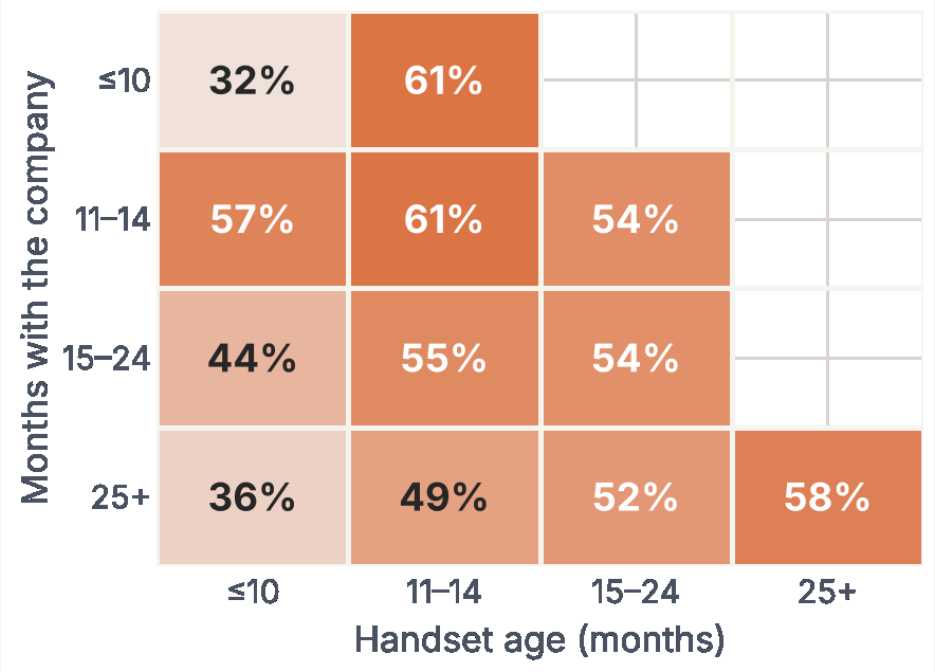
Customers with Older Handsets Show Higher Churn Rates

The data shows higher churn rates among customers with older handsets, including within some customer-tenure groups. This makes handset age a useful signal for customer-retention analysis. However, the analysis does not prove that an older handset causes churn. The proposed pilot will test whether offering a handset upgrade improves retention compared with a control group.

Churn jumps from about 38% to 62% once a phone passes 10 months



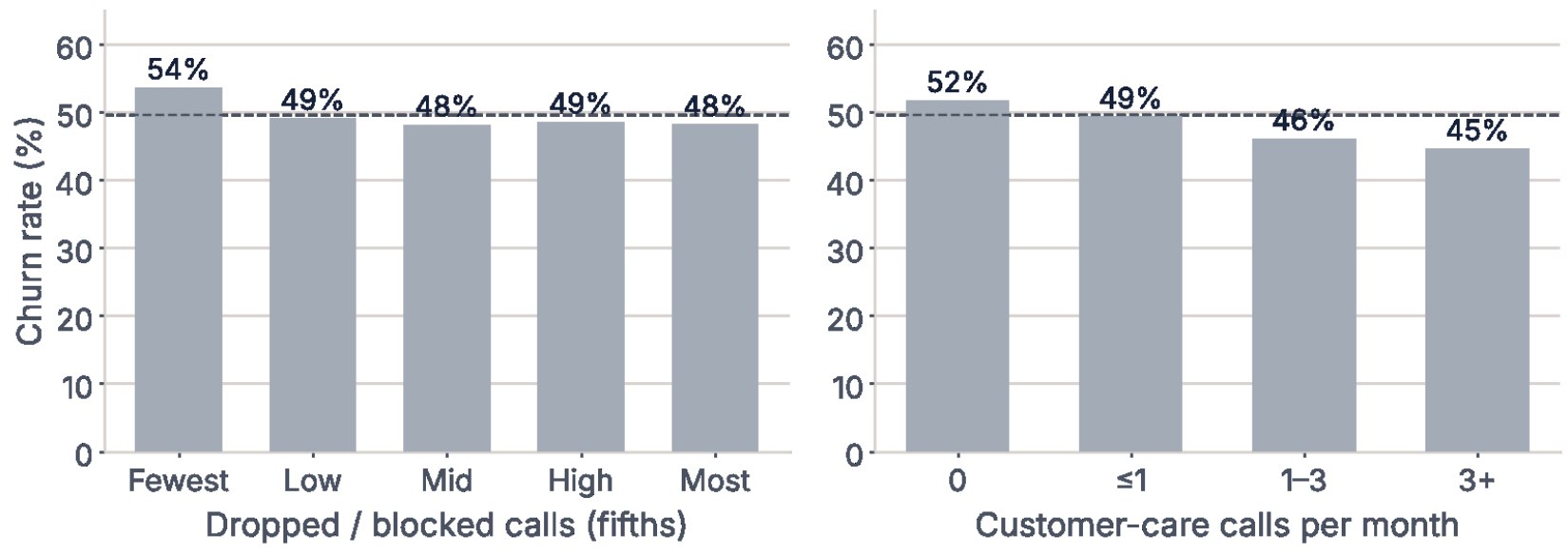
Same tenure, newer phone, lower churn: 36% vs 58% after 25+ months



Source: Company A data, n = 100,000. Cells with fewer than 500 customers are left blank.

FINDING 3 · WHAT IT IS NOT

No sign that dropped calls or care calls drive churn here



Customers with more dropped calls, or more calls to customer care, do **not** leave more often. Heavy users generate more of both, and heavy users tend to stay. These are counts, not drop rates, so the network may still matter; it is simply not where this data points. Churn is 55% for the lightest users vs 46% for the heaviest.

Source: Company A data, n = 100,000. Dashed line = 49.6% average.

PROBLEM DEFINITION

The question we set out to answer

> *"Which customers will leave in the next one to two months, so we can make them the right offer before they decide?"*

Choice	What we used, and why
Target	churn: the customer left 31–60 days after the observation date
ML task	Binary classification that gives every customer a risk score
Main metric	ROC-AUC: how well the model ranks future leavers above stayers
Also reported	Accuracy, precision, recall, F1, and churners caught in the top 10–30%
Who acts on it	The retention team, with a fresh ranked list every month

Ethnicity was left out on purpose: who receives an offer must not depend on it. Other household fields add only +0.005 AUC and could also be dropped in production.

THE MODEL

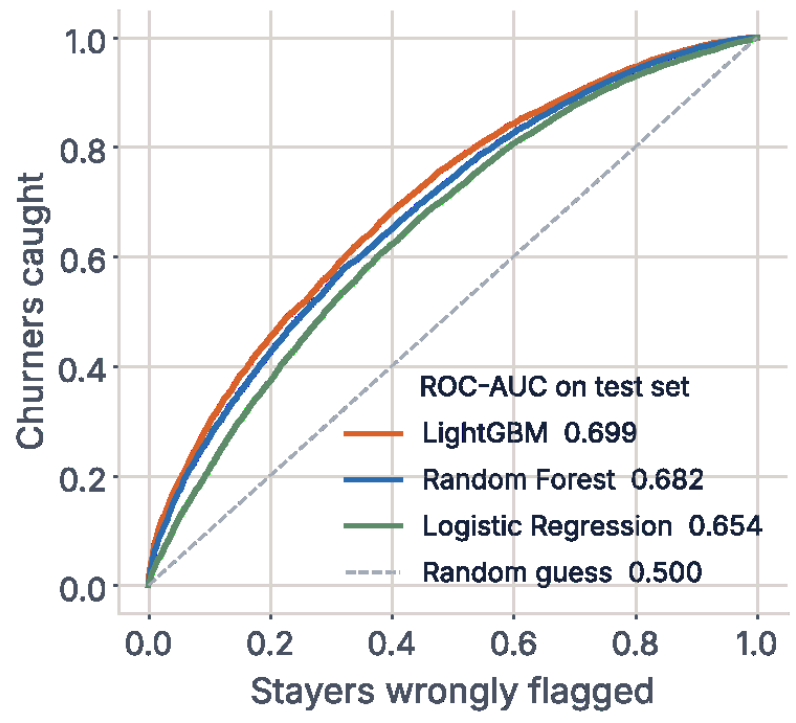
LightGBM: ROC-AUC 0.699 on 30,000 customers it never saw

Model (test set)	ROC-AUC	Accuracy	F1
LightGBM (chosen)	0.699	0.640	0.639
Random Forest	0.682	0.625	0.630
Logistic Regression	0.654	0.611	0.621
Baseline (always “stay”)	0.500	0.504	0.000

Why LightGBM: hundreds of small decision trees, each fixing the last one’s errors; it handles missing values well.

How it was tested: chosen on cross-validation (0.698 ± 0.002 across folds), then scored once on 30,000 held-back customers.

Inputs: 102 features; the 8 built from our analysis (handset age, tenure window, usage trend…) add +0.004 AUC.

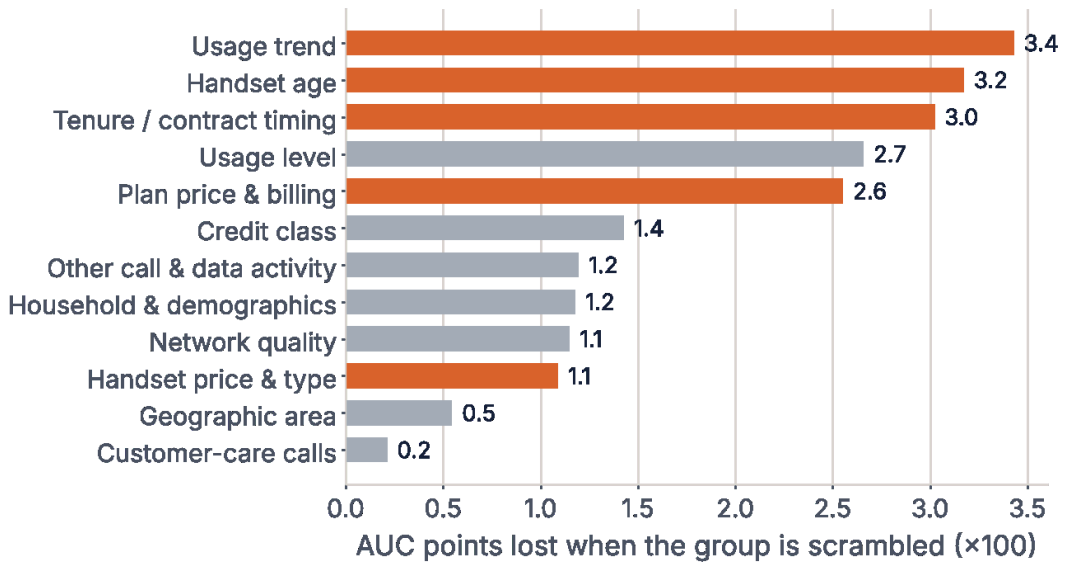


Accuracy, precision (0.635), recall (0.643) and F1 at a 0.5 threshold. Settings fixed before testing. Notebook section 6.

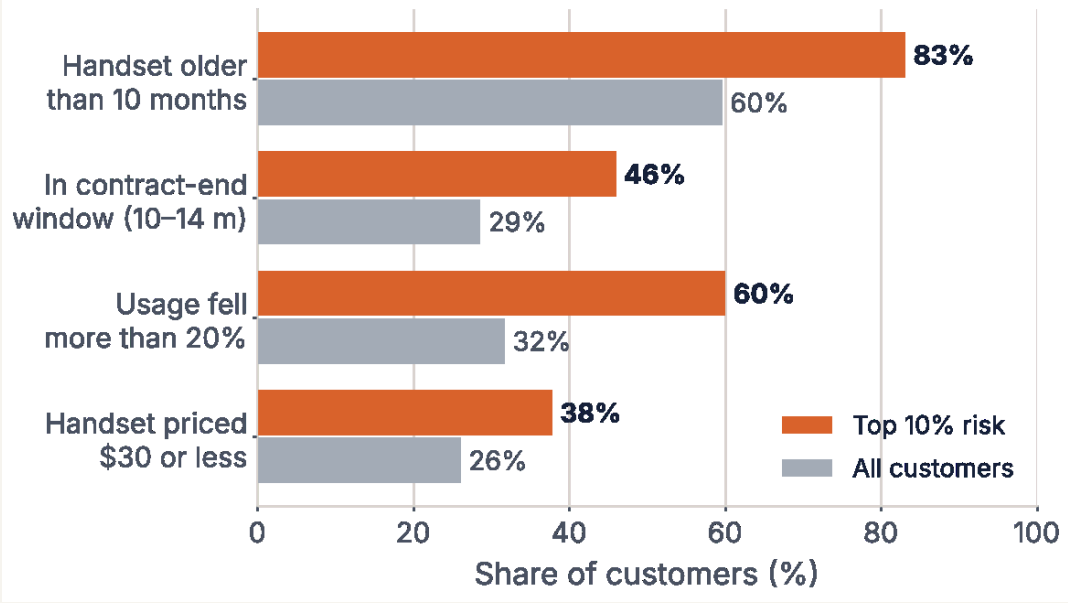
WHAT THE MODEL LEARNED

The model picked up the same patterns as our analysis

Strongest signals: usage trend, handset age and contract timing



83% of the riskiest 10% carry a handset older than 10 months

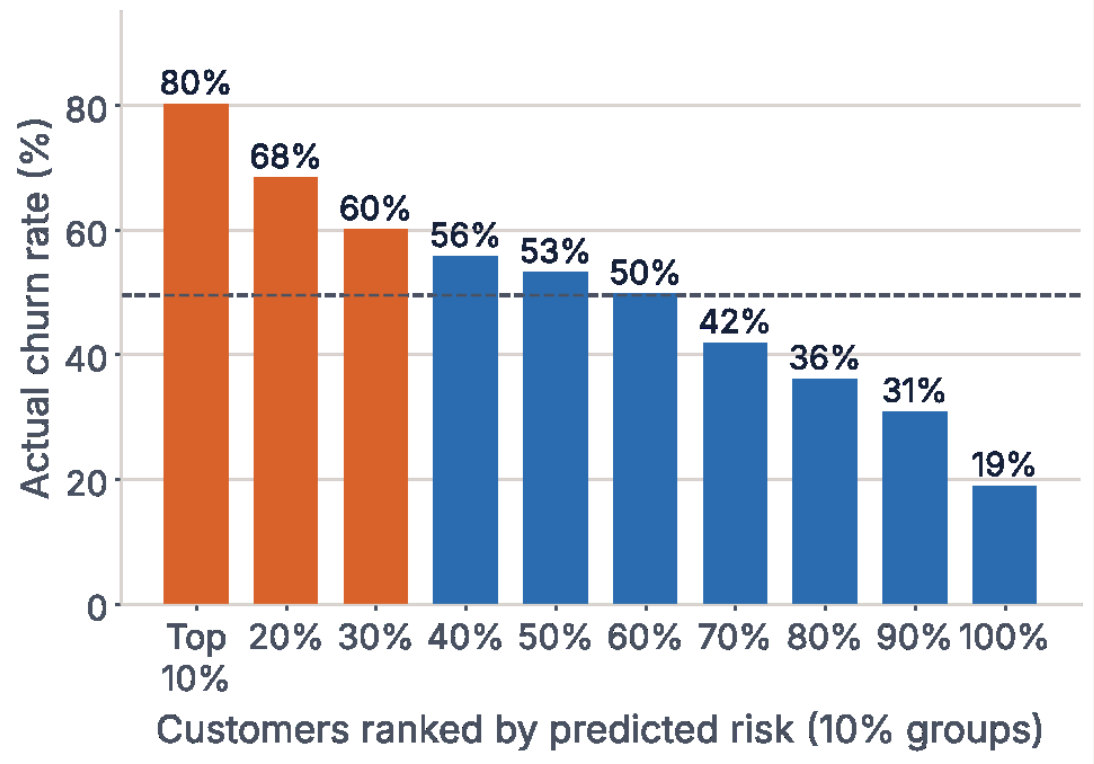


Orange = levers Company A can act on. Importance = AUC lost when a group is scrambled: reliance, not cause.

Source: LightGBM on the 30,000-customer test set; grouped permutation importance on a 10,000 sample.

WHY THE MODEL IS WORTH USING

A moderate score still pays, because it ranks the right customers first



80%
of the riskiest 10% leave in the sample: 1.6 times the average

42%
of all leavers sit in the riskiest 30% of customers

19%
churn in the safest 10%: no reason to spend money there

Source: LightGBM on 30,000 test customers (balanced sample rates). Dashed line = 49.6% average.

THE PROPOSAL · TO BE PILOTED

Smart Upgrade: A Targeted Handset Offer to Test

01

Score

Every month, the model gives each customer a risk score.

02

Select

Pick the riskiest ~0.7%, first those in months 9–10 (timing to test).

03

Offer

A new handset on EMI with a ~$30 subsidy, tied to a 12-month renewal.

04

Measure

Keep a random control group, so savings are proven, not assumed.

> **Why a phone upgrade?**
>
> 83% of the riskiest customers carry a handset older than 10 months, and Indian buyers pay a phone off in about 10 months [6]. If phone age is part of the reason they leave, the offer reaches them at the right moment. The pilot tests whether it does.

Sources: Company A data and model (slides 5–11); [6] Counterpoint, Aug 2026

BUSINESS IMPACT · AN ESTIMATE, NOT A PROMISE

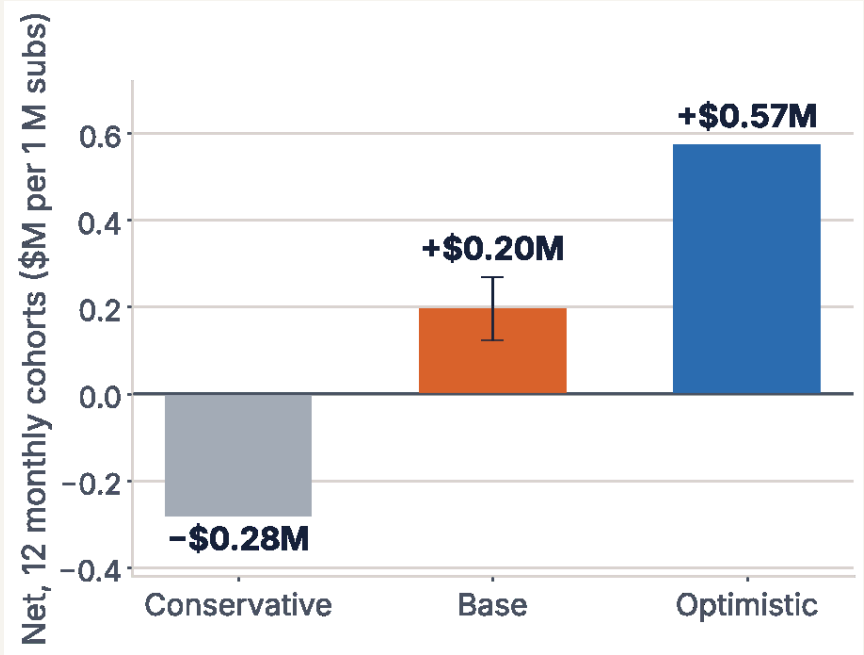
Base case: +$0.2 M per 1 M subscribers, 12 campaigns

Scenario	Conservative	Base	Optimistic
Churners the offer saves	15%	25%	30%
Months a saved customer stays	12	18	24
Customers saved / month	~90	~150	~180
Net, 12 monthly cohorts	−$0.28M	+$0.20M	+$0.57M

Each month ~6,900 customers are contacted (riskiest 0.7%). At the assumed 1.1% churn, about 8.6% of them would leave, against 1.1% at random. **The same offer sent to everyone would lose about $48.7M.**

Also assumed: 1.1% assumed monthly churn rate, used as an illustrative business-scenario input. The value is informed by public MNP request volumes but is not a measured churn rate for Company A. [2]· $40 monthly revenue (median of the top-risk group) · 50% margin · $2 per contact, $30 subsidy per upgrade · 10–15% of stayers also take the subsidy.

Expected lifetime value of 12 cohorts, test set; bar = 90% sampling interval

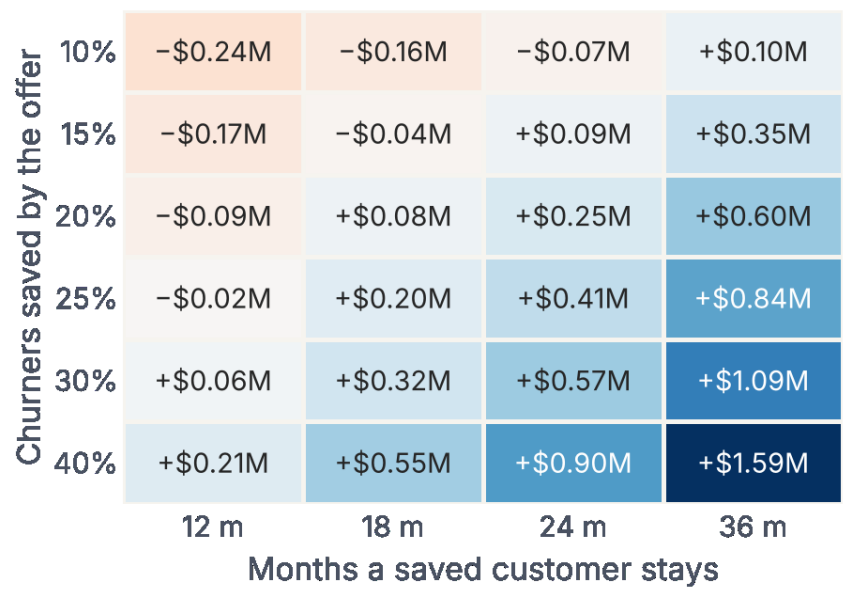


Money in the dataset's units (shown as $). Threshold chosen on training data, tested once. Notebook 7.4–7.7.

RISKS AND NEXT STEPS

Start with a pilot that tests the assumptions

Net, 12 monthly cohorts ($M per 1 M subs), other assumptions



Positive only if the offer saves enough leavers and they stay past the 12-month renewal.

The value rests on assumptions

The conservative case loses $0.28M over 12 cohorts, and the base case turns negative if 20% of stayers take the subsidy. Nothing scales before the pilot.

Contact can backfire

In a telecom field test, proactive plan advice raised churn from 6% to 10% [9].

Riskiest is not always most persuadable

Targeting by expected response can beat targeting by risk [10]: an uplift model, but only after the pilot.

3-month pilot: offer vs random control group

Measure: extra customers kept · offer uptake · cost per customer kept · net value · complaints. Scale up only if results land in the blue area.

Sources: [9] Knowledge at Wharton, 2015 · [10] Ascarza, Journal of Marketing Research, 2018 · notebook sections 7.5–7.8

REFERENCES

Sources

[1] TelecomTalk (28 Sep 2026). India wireless subscriber base, August 2026 (TRAI data).

[2] Press Information Bureau (25 Jun 2026). Telecom subscription data, May 2026; and TelecomTalk (28 Jun 2026).

[3] TelecomTalk (13 Aug 2026). Airtel, Reliance Jio, Vodafone Idea ARPU in Q1 FY27.

[4] TelecomTalk (4 Aug 2026). Bharti Airtel ARPU Rs 264, postpaid base 30 million, Q1 FY27.

[5] Counterpoint Research (3 Jul 2026). Rising prices to push financing to 42% of India's smartphone sales in 2026.

[6] Counterpoint Research (10 Aug 2026). Average smartphone financing tenure reaches 10 months in Q2 2026.

[7] Multibagg AI (22 Apr 2026). Indian telecom tariff hike 2026: analysts' estimates.

[8] Gallo, A. (2014). The Value of Keeping the Right Customers. Harvard Business Review.

[9] Knowledge at Wharton (2015). Why retention plans don't work (Ascarza, Iyengar & Schleicher).

[10] Ascarza, E. (2018). Retention Futility: Targeting High-Risk Customers Might Be Ineffective. Journal of Marketing Research.

[11] GCI World 2026 September: Final Assignment README, Dataset Overview and tutorial notebook (Matsuo-Iwasawa Lab, UTokyo).

[12] Generative AI (chatgpt) used for coding support and editing; https://chatgpt.com/share/6ad5a390-b994-83e8-8c19-6d72c752c123

Full URLs are listed in the Omnicampus reference field and at the end of the submitted notebook.